

Mathematical Foundations of Machine Learning

Assignment 8 Solutions

Question 1 (15 points)

Consider a one-layer neural network with input $\mathbf{x} \in \mathbb{R}^2$, weight vector $\mathbf{w} \in \mathbb{R}^2$, and bias $b \in \mathbb{R}$. Define

$$z = \mathbf{w}^\top \mathbf{x} + b, \quad \hat{y} = \tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}},$$

and the loss

$$L(\mathbf{w}, b) = \frac{1}{2}(y - \hat{y})^2,$$

where $y \in \mathbb{R}$ is the target value.

(a) (10 points) Derive $\frac{dL}{d\mathbf{w}}$ and $\frac{dL}{db}$.

(b) (5 points) Evaluate z , $\hat{y}$, L , $\frac{dL}{d\mathbf{w}}$, and $\frac{dL}{db}$ at

$$\mathbf{x} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad b = 0, \quad y = 0.$$

Solution.

(a) By the chain rule,

$$\frac{dL}{d\hat{y}} = \hat{y} - y.$$

Since $\hat{y} = \tanh(z)$, we have

$$\frac{d\hat{y}}{dz} = 1 - \tanh^2(z) = 1 - \hat{y}^2.$$

Also,

$$\frac{dz}{d\mathbf{w}} = \mathbf{x}^\top, \quad \frac{dz}{db} = 1.$$

Therefore,

$$\frac{dL}{d\mathbf{w}} = \frac{dL}{d\hat{y}} \frac{d\hat{y}}{dz} \frac{dz}{d\mathbf{w}} = (\hat{y} - y)(1 - \hat{y}^2)\mathbf{x}^\top,$$

and

$$\frac{dL}{db} = \frac{dL}{d\hat{y}} \frac{d\hat{y}}{dz} \frac{dz}{db} = (\hat{y} - y)(1 - \hat{y}^2).$$

(b) First,

$$z = \mathbf{w}^\top \mathbf{x} + b = \begin{bmatrix} 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} + 0 = 1.$$

Hence

$$\hat{y} = \tanh(1).$$

Since $y = 0$,

$$L = \frac{1}{2} \tanh^2(1).$$

Using part (a),

$$\frac{dL}{dw} = \tanh(1)(1 - \tanh^2(1)) \begin{bmatrix} 1 & -1 \end{bmatrix},$$

and

$$\frac{dL}{db} = \tanh(1)(1 - \tanh^2(1)).$$

Question 2 (20 points)

Consider a two-layer neural network

$$\begin{aligned} \mathbf{f}_0 &= \mathbf{x}, & \mathbf{z}_1 &= \mathbf{A}_0 \mathbf{f}_0 + \mathbf{b}_0, & \mathbf{f}_1 &= \sigma_1(\mathbf{z}_1), \\ \mathbf{z}_2 &= \mathbf{A}_1 \mathbf{f}_1 + \mathbf{b}_1, & \mathbf{f}_2 &= \sigma_2(\mathbf{z}_2), \end{aligned}$$

where

$$\mathbf{x} \in \mathbb{R}^n, \quad \mathbf{A}_0 \in \mathbb{R}^{p \times n}, \quad \mathbf{b}_0 \in \mathbb{R}^p, \quad \mathbf{A}_1 \in \mathbb{R}^{m \times p}, \quad \mathbf{b}_1 \in \mathbb{R}^m,$$

and σ_1, σ_2 are differentiable and applied componentwise. Let the loss be

$$L = \frac{1}{2} \|\mathbf{f}_2 - \mathbf{y}\|_2^2, \quad \mathbf{y} \in \mathbb{R}^m.$$

Define:

$$\begin{aligned} \frac{\partial \mathbf{f}_1}{\partial \mathbf{z}_1} &= \text{Diag} \left(\frac{\partial \sigma_1(z_{1,1})}{\partial z_{1,1}}, \dots, \frac{\partial \sigma_1(z_{1,p})}{\partial z_{1,p}} \right), \\ \frac{\partial \mathbf{f}_2}{\partial \mathbf{z}_2} &= \text{Diag} \left(\frac{\partial \sigma_2(z_{2,1})}{\partial z_{2,1}}, \dots, \frac{\partial \sigma_2(z_{2,m})}{\partial z_{2,m}} \right), \end{aligned}$$

where these derivatives are taken element-wise.

(a) (5 points) Show that

$$\frac{\partial L}{\partial \mathbf{z}_2} = \frac{\partial L}{\partial \mathbf{f}_2} \frac{\partial \mathbf{f}_2}{\partial \mathbf{z}_2} = (\mathbf{f}_2 - \mathbf{y})^\top \frac{\partial \mathbf{f}_2}{\partial \mathbf{z}_2}.$$

(b) (6 points) Using $\frac{\partial L}{\partial \mathbf{z}_2}$, derive formulas for

$$\frac{\partial L}{\partial \mathbf{A}_1} \quad \text{and} \quad \frac{\partial L}{\partial \mathbf{b}_1}.$$

(c) (6 points) Show that

$$\frac{\partial L}{\partial \mathbf{z}_1} = \frac{\partial L}{\partial \mathbf{z}_2} \frac{\partial \mathbf{z}_2}{\partial \mathbf{f}_1} \frac{\partial \mathbf{f}_1}{\partial \mathbf{z}_1} = \frac{\partial L}{\partial \mathbf{z}_2} \mathbf{A}_1 \frac{\partial \mathbf{f}_1}{\partial \mathbf{z}_1},$$

and derive formulas for

$$\frac{\partial L}{\partial \mathbf{A}_0} \quad \text{and} \quad \frac{\partial L}{\partial \mathbf{b}_0}.$$

(d) (3 points) State the dimensions of

$$\mathbf{z}_1, \mathbf{f}_1, \mathbf{z}_2, \mathbf{f}_2, \frac{\partial L}{\partial \mathbf{z}_1}, \frac{\partial L}{\partial \mathbf{z}_2}, \frac{\partial \mathbf{f}_1}{\partial \mathbf{z}_1}, \frac{\partial \mathbf{f}_2}{\partial \mathbf{z}_2}, \frac{\partial L}{\partial \mathbf{A}_0}, \frac{\partial L}{\partial \mathbf{b}_0}, \frac{\partial L}{\partial \mathbf{A}_1}, \frac{\partial L}{\partial \mathbf{b}_1}.$$

Solution.

(a) Since

$$L = \frac{1}{2} \|\mathbf{f}_2 - \mathbf{y}\|_2^2 = \frac{1}{2} \sum_{i=1}^m (f_{2,i} - y_i)^2,$$

we have

$$\frac{\partial L}{\partial \mathbf{f}_2} = (\mathbf{f}_2 - \mathbf{y})^\top.$$

Because $\mathbf{f}_2$ is a vector-valued function of $\mathbf{z}_2$, the chain rule gives

$$\frac{\partial L}{\partial \mathbf{z}_2} = \frac{\partial L}{\partial \mathbf{f}_2} \frac{\partial \mathbf{f}_2}{\partial \mathbf{z}_2} = (\mathbf{f}_2 - \mathbf{y})^\top \frac{\partial \mathbf{f}_2}{\partial \mathbf{z}_2}.$$

(b) Since

$$\mathbf{z}_2 = \mathbf{A}_1 \mathbf{f}_1 + \mathbf{b}_1,$$

its differential is

$$d\mathbf{z}_2 = d\mathbf{A}_1 \mathbf{f}_1 + d\mathbf{b}_1.$$

Therefore

$$dL = \frac{\partial L}{\partial \mathbf{z}_2} d\mathbf{z}_2 = \frac{\partial L}{\partial \mathbf{z}_2} d\mathbf{A}_1 \mathbf{f}_1 + \frac{\partial L}{\partial \mathbf{z}_2} d\mathbf{b}_1.$$

For the matrix term,

$$\frac{\partial L}{\partial \mathbf{z}_2} d\mathbf{A}_1 \mathbf{f}_1 = \text{tr} \left(\frac{\partial L}{\partial \mathbf{z}_2} d\mathbf{A}_1 \mathbf{f}_1 \right) = \text{tr} \left(\mathbf{f}_1 \frac{\partial L}{\partial \mathbf{z}_2} d\mathbf{A}_1 \right) = \text{tr} \left(\left(\left(\frac{\partial L}{\partial \mathbf{z}_2} \right)^\top \mathbf{f}_1^\top \right)^\top d\mathbf{A}_1 \right).$$

Hence

$$\frac{\partial L}{\partial \mathbf{A}_1} = \left(\frac{\partial L}{\partial \mathbf{z}_2} \right)^\top \mathbf{f}_1^\top.$$

Also,

$$dL = \frac{\partial L}{\partial \mathbf{z}_2} d\mathbf{b}_1 \implies \frac{\partial L}{\partial \mathbf{b}_1} = \frac{\partial L}{\partial \mathbf{z}_2}.$$

(c) First, since

$$\mathbf{z}_2 = \mathbf{A}_1 \mathbf{f}_1 + \mathbf{b}_1,$$

we have

$$\frac{\partial \mathbf{z}_2}{\partial \mathbf{f}_1} = \mathbf{A}_1.$$

Therefore, by the chain rule,

$$\frac{\partial L}{\partial \mathbf{f}_1} = \frac{\partial L}{\partial \mathbf{z}_2} \frac{\partial \mathbf{z}_2}{\partial \mathbf{f}_1} = \frac{\partial L}{\partial \mathbf{z}_2} \mathbf{A}_1.$$

Because $\mathbf{f}_1 = \sigma_1(\mathbf{z}_1)$, we obtain

$$\frac{\partial L}{\partial \mathbf{z}_1} = \frac{\partial L}{\partial \mathbf{f}_1} \frac{\partial \mathbf{f}_1}{\partial \mathbf{z}_1} = \frac{\partial L}{\partial \mathbf{z}_2} \mathbf{A}_1 \frac{\partial \mathbf{f}_1}{\partial \mathbf{z}_1}.$$

Now

$$\mathbf{z}_1 = \mathbf{A}_0 \mathbf{x} + \mathbf{b}_0, \quad d\mathbf{z}_1 = d\mathbf{A}_0 \mathbf{x} + d\mathbf{b}_0.$$

Therefore

$$dL = \frac{\partial L}{\partial \mathbf{z}_1} d\mathbf{z}_1 = \frac{\partial L}{\partial \mathbf{z}_1} d\mathbf{A}_0 \mathbf{x} + \frac{\partial L}{\partial \mathbf{z}_1} d\mathbf{b}_0.$$

Using the same trace argument as in part (b),

$$\frac{\partial L}{\partial \mathbf{A}_0} = \left(\frac{\partial L}{\partial \mathbf{z}_1} \right)^\top \mathbf{x}^\top, \quad \frac{\partial L}{\partial \mathbf{b}_0} = \frac{\partial L}{\partial \mathbf{z}_1}.$$

(d) The dimensions are

$$\begin{aligned}
\mathbf{z}_1 &\in \mathbb{R}^{p \times 1}, & \mathbf{f}_1 &\in \mathbb{R}^{p \times 1}, & \mathbf{z}_2 &\in \mathbb{R}^{m \times 1}, & \mathbf{f}_2 &\in \mathbb{R}^{m \times 1}, \\
\frac{\partial L}{\partial \mathbf{z}_1} &\in \mathbb{R}^{1 \times p}, & \frac{\partial L}{\partial \mathbf{z}_2} &\in \mathbb{R}^{1 \times m}, \\
\frac{\partial \mathbf{f}_1}{\partial \mathbf{z}_1} &\in \mathbb{R}^{p \times p}, & \frac{\partial \mathbf{f}_2}{\partial \mathbf{z}_2} &\in \mathbb{R}^{m \times m}, \\
\frac{\partial L}{\partial \mathbf{A}_0} &\in \mathbb{R}^{p \times n}, & \frac{\partial L}{\partial \mathbf{b}_0} &\in \mathbb{R}^{1 \times p}, \\
\frac{\partial L}{\partial \mathbf{A}_1} &\in \mathbb{R}^{m \times p}, & \frac{\partial L}{\partial \mathbf{b}_1} &\in \mathbb{R}^{1 \times m}.
\end{aligned}$$

Question 3 (20 points)

Linear regression seeks a vector $\boldsymbol{\theta} \in \mathbb{R}^d$ that minimizes

$$J(\boldsymbol{\theta}) = \frac{1}{N} (\mathbf{X}\boldsymbol{\theta} - \mathbf{y})^\top (\mathbf{X}\boldsymbol{\theta} - \mathbf{y}),$$

where $\mathbf{X} \in \mathbb{R}^{N \times d}$ and $\mathbf{y} \in \mathbb{R}^N$.

- (a) (6 points) Derive $\frac{dJ}{d\boldsymbol{\theta}}$ and $\frac{d^2J}{d\boldsymbol{\theta}^2}$. Show that the Hessian is positive semidefinite, and state a condition under which it is positive definite.
- (b) (6 points) Let $\boldsymbol{\theta}_0 \in \mathbb{R}^d$ be arbitrary, and define

$$\boldsymbol{\delta} = \boldsymbol{\theta} - \boldsymbol{\theta}_0.$$

Write the second-degree Taylor polynomial of J at $\boldsymbol{\theta}_0$ in terms of $J(\boldsymbol{\theta}_0)$, $\frac{dJ(\boldsymbol{\theta}_0)}{d\boldsymbol{\theta}}$, $\frac{d^2J(\boldsymbol{\theta}_0)}{d\boldsymbol{\theta}^2}$, and $\boldsymbol{\delta}$. Then show that this Taylor polynomial is exact, i.e., it equals $J(\boldsymbol{\theta})$ for every $\boldsymbol{\theta} \in \mathbb{R}^d$.

- (c) (8 points) Assume $\mathbf{X}^\top \mathbf{X}$ is invertible. Derive the unique stationary point

$$\boldsymbol{\theta}^* = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}.$$

Then use part (b) with $\boldsymbol{\theta}_0 = \boldsymbol{\theta}^*$ to show that for every $\boldsymbol{\theta} \in \mathbb{R}^d$,

$$J(\boldsymbol{\theta}) = J(\boldsymbol{\theta}^*) + \frac{1}{2} (\boldsymbol{\theta} - \boldsymbol{\theta}^*)^\top \frac{d^2J(\boldsymbol{\theta}^*)}{d\boldsymbol{\theta}^2} (\boldsymbol{\theta} - \boldsymbol{\theta}^*).$$

Deduce that $\boldsymbol{\theta}^*$ is the unique global minimizer of J .

Solution.

- (a) First expand

$$\begin{aligned}
J(\boldsymbol{\theta}) &= \frac{1}{N} (\mathbf{X}\boldsymbol{\theta} - \mathbf{y})^\top (\mathbf{X}\boldsymbol{\theta} - \mathbf{y}) \\
&= \frac{1}{N} (\boldsymbol{\theta}^\top \mathbf{X}^\top \mathbf{X} \boldsymbol{\theta} - 2\mathbf{y}^\top \mathbf{X} \boldsymbol{\theta} + \mathbf{y}^\top \mathbf{y}).
\end{aligned}$$

Since $\mathbf{X}^\top \mathbf{X}$ is symmetric,

$$\frac{d}{d\boldsymbol{\theta}} (\boldsymbol{\theta}^\top \mathbf{X}^\top \mathbf{X} \boldsymbol{\theta}) = 2\boldsymbol{\theta}^\top \mathbf{X}^\top \mathbf{X}, \quad \frac{d}{d\boldsymbol{\theta}} (-2\mathbf{y}^\top \mathbf{X} \boldsymbol{\theta}) = -2\mathbf{y}^\top \mathbf{X}.$$

Therefore

$$\frac{dJ}{d\boldsymbol{\theta}} = \frac{2}{N}(\boldsymbol{\theta}^\top X^\top X - \mathbf{y}^\top X) = \frac{2}{N}(X\boldsymbol{\theta} - \mathbf{y})^\top X.$$

Differentiating once more gives

$$\frac{d^2 J}{d\boldsymbol{\theta}^2} = \frac{2}{N}X^\top X.$$

To show it is positive semidefinite, let $\mathbf{z} \in \mathbb{R}^d$. Then

$$\mathbf{z}^\top \frac{d^2 J}{d\boldsymbol{\theta}^2} \mathbf{z} = \frac{2}{N} \mathbf{z}^\top X^\top X \mathbf{z} = \frac{2}{N} \|X\mathbf{z}\|_2^2 \geq 0.$$

Hence the Hessian is positive semidefinite.

It is positive definite if and only if X has full column rank.

(b) The second-degree Taylor polynomial of J at $\boldsymbol{\theta}_0$ is

$$T_2(\boldsymbol{\theta}) = J(\boldsymbol{\theta}_0) + \frac{dJ(\boldsymbol{\theta}_0)}{d\boldsymbol{\theta}} \boldsymbol{\delta} + \frac{1}{2} \boldsymbol{\delta}^\top \frac{d^2 J(\boldsymbol{\theta}_0)}{d\boldsymbol{\theta}^2} \boldsymbol{\delta}, \quad \boldsymbol{\delta} = \boldsymbol{\theta} - \boldsymbol{\theta}_0.$$

Now write $\boldsymbol{\theta} = \boldsymbol{\theta}_0 + \boldsymbol{\delta}$. Then

$$X\boldsymbol{\theta} - \mathbf{y} = X\boldsymbol{\theta}_0 - \mathbf{y} + X\boldsymbol{\delta}.$$

Hence

$$\begin{aligned} J(\boldsymbol{\theta}) &= \frac{1}{N}(X\boldsymbol{\theta}_0 - \mathbf{y} + X\boldsymbol{\delta})^\top (X\boldsymbol{\theta}_0 - \mathbf{y} + X\boldsymbol{\delta}) \\ &= \frac{1}{N}(X\boldsymbol{\theta}_0 - \mathbf{y})^\top (X\boldsymbol{\theta}_0 - \mathbf{y}) + \frac{2}{N}(X\boldsymbol{\theta}_0 - \mathbf{y})^\top X\boldsymbol{\delta} + \frac{1}{N}\boldsymbol{\delta}^\top X^\top X\boldsymbol{\delta}. \end{aligned}$$

The first term is $J(\boldsymbol{\theta}_0)$, the second term is

$$\frac{dJ(\boldsymbol{\theta}_0)}{d\boldsymbol{\theta}} \boldsymbol{\delta},$$

and the third term is

$$\frac{1}{2} \boldsymbol{\delta}^\top \frac{d^2 J(\boldsymbol{\theta}_0)}{d\boldsymbol{\theta}^2} \boldsymbol{\delta}.$$

Therefore

$$J(\boldsymbol{\theta}) = J(\boldsymbol{\theta}_0) + \frac{dJ(\boldsymbol{\theta}_0)}{d\boldsymbol{\theta}} \boldsymbol{\delta} + \frac{1}{2} \boldsymbol{\delta}^\top \frac{d^2 J(\boldsymbol{\theta}_0)}{d\boldsymbol{\theta}^2} \boldsymbol{\delta}.$$

So the second-degree Taylor polynomial is exact.

(c) A stationary point satisfies

$$\frac{dJ}{d\boldsymbol{\theta}} = \mathbf{0}^\top.$$

Using part (a),

$$\frac{2}{N}(X\boldsymbol{\theta} - \mathbf{y})^\top X = \mathbf{0}^\top.$$

Taking transpose gives

$$X^\top (X\boldsymbol{\theta} - \mathbf{y}) = \mathbf{0},$$

so

$$X^\top X\boldsymbol{\theta} = X^\top \mathbf{y}.$$

Since $X^\top X$ is invertible, the unique stationary point is

$$\boldsymbol{\theta}^* = (X^\top X)^{-1} X^\top \mathbf{y}.$$

Now apply part (b) with $\boldsymbol{\theta}_0 = \boldsymbol{\theta}^*$. Since

$$\frac{dJ(\boldsymbol{\theta}^*)}{d\boldsymbol{\theta}} = \mathbf{0}^\top,$$

the linear term vanishes, and we obtain

$$J(\boldsymbol{\theta}) = J(\boldsymbol{\theta}^*) + \frac{1}{2}(\boldsymbol{\theta} - \boldsymbol{\theta}^*)^\top \frac{d^2 J(\boldsymbol{\theta}^*)}{d\boldsymbol{\theta}^2} (\boldsymbol{\theta} - \boldsymbol{\theta}^*).$$

Because $X^\top X$ is invertible, the Hessian

$$\frac{d^2 J(\boldsymbol{\theta}^*)}{d\boldsymbol{\theta}^2} = \frac{2}{N} X^\top X$$

is positive definite. Therefore, for every $\boldsymbol{\theta} \neq \boldsymbol{\theta}^*$,

$$(\boldsymbol{\theta} - \boldsymbol{\theta}^*)^\top \frac{d^2 J(\boldsymbol{\theta}^*)}{d\boldsymbol{\theta}^2} (\boldsymbol{\theta} - \boldsymbol{\theta}^*) > 0.$$

Hence

$$J(\boldsymbol{\theta}) > J(\boldsymbol{\theta}^*) \quad \text{for all } \boldsymbol{\theta} \neq \boldsymbol{\theta}^*.$$

So $\boldsymbol{\theta}^*$ is the unique global minimizer of J .

Question 4 (15 points)

Consider the function

$$f(x, y) = x^3 + xy + y^2.$$

- (a) (6 points) Compute $\frac{df(x, y)}{d(x, y)}$.
- (b) (6 points) Compute the Hessian matrix $\nabla_{x, y}^2 f(x, y)$ and verify that the mixed second-order partial derivatives agree.
- (c) (3 points) Evaluate $\nabla_{x, y}^2 f(0, 0)$ and determine whether the local quadratic term is positive definite, negative definite, or indefinite at $(0, 0)$.

Solution.

- (a) The first-order partial derivatives are

$$\frac{\partial f}{\partial x} = 3x^2 + y, \quad \frac{\partial f}{\partial y} = x + 2y.$$

Therefore

$$\frac{df(x, y)}{d(x, y)} = \begin{bmatrix} 3x^2 + y & x + 2y \end{bmatrix}.$$

- (b) The second-order partial derivatives are

$$\frac{\partial^2 f}{\partial x^2} = 6x, \quad \frac{\partial^2 f}{\partial x \partial y} = 1, \quad \frac{\partial^2 f}{\partial y \partial x} = 1, \quad \frac{\partial^2 f}{\partial y^2} = 2.$$

Hence

$$\nabla_{x, y}^2 f(x, y) = \begin{bmatrix} 6x & 1 \\ 1 & 2 \end{bmatrix}.$$

The mixed second-order partial derivatives agree.

(c) At $(0, 0)$,

$$\nabla_{x,y}^2 f(0, 0) = \begin{bmatrix} 0 & 1 \\ 1 & 2 \end{bmatrix}.$$

Its determinant is

$$-1 < 0,$$

so the Hessian is indefinite. Therefore the local quadratic term

$$\frac{1}{2} \delta^\top \nabla_{x,y}^2 f(0, 0) \delta$$

is also indefinite.

Question 5 (10 points)

Consider the function

$$f(x, y) = e^{x-y} + xy.$$

- (a) (3 points) Find the linearization of f at the point $(0, 0)$.
- (b) (4 points) Find the second-degree Taylor polynomial of f at the point $(0, 0)$.
- (c) (3 points) Use both approximations from parts (a) and (b) to estimate $f(0.1, 0.05)$. Also compute the exact value, and state which approximation is more accurate.

Solution.

(a) First,

$$f(0, 0) = 1.$$

The first-order partial derivatives are

$$\frac{\partial f}{\partial x} = e^{x-y} + y, \quad \frac{\partial f}{\partial y} = -e^{x-y} + x.$$

Hence

$$\frac{df(0, 0)}{d(x, y)} = \begin{bmatrix} 1 & -1 \end{bmatrix}.$$

Therefore the linearization at $(0, 0)$ is

$$T_1(x, y) = f(0, 0) + \frac{df(0, 0)}{d(x, y)} \begin{bmatrix} x \\ y \end{bmatrix} = 1 + x - y.$$

(b) The second-order partial derivatives are

$$\frac{\partial^2 f}{\partial x^2} = e^{x-y}, \quad \frac{\partial^2 f}{\partial x \partial y} = -e^{x-y} + 1, \quad \frac{\partial^2 f}{\partial y^2} = e^{x-y}.$$

Thus

$$\nabla_{x,y}^2 f(0, 0) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

The second-degree Taylor polynomial at $(0, 0)$ is

$$T_2(x, y) = f(0, 0) + \frac{df(0, 0)}{d(x, y)} \begin{bmatrix} x \\ y \end{bmatrix} + \frac{1}{2} \begin{bmatrix} x & y \end{bmatrix} \nabla_{x,y}^2 f(0, 0) \begin{bmatrix} x \\ y \end{bmatrix}.$$

Therefore

$$T_2(x, y) = 1 + x - y + \frac{1}{2}(x^2 + y^2).$$

(c) Using the linearization,

$$T_1(0.1, 0.05) = 1.05.$$

Using the second-degree Taylor polynomial,

$$T_2(0.1, 0.05) = 1.05625.$$

The exact value is

$$f(0.1, 0.05) = e^{0.05} + 0.005 \approx 1.05627.$$

Hence the second-degree Taylor approximation is more accurate.

Question 6 (20 points)

Consider the function

$$f(x, y) = e^x + xy^2.$$

Let $(x_0, y_0) = (0, 0)$, and define

$$\boldsymbol{\delta} = \begin{bmatrix} x \\ y \end{bmatrix}.$$

Recall that the third-degree multivariate Taylor polynomial at $(0, 0)$ is

$$T_3(x, y) = f(0, 0) + \frac{df(0, 0)}{d(x, y)}\boldsymbol{\delta} + \frac{1}{2}\boldsymbol{\delta}^\top \nabla_{x,y}^2 f(0, 0)\boldsymbol{\delta} + \frac{1}{6}\nabla_{x,y}^3 f(0, 0)\boldsymbol{\delta}^3,$$

where $\boldsymbol{\delta}^3 = \boldsymbol{\delta} \otimes \boldsymbol{\delta} \otimes \boldsymbol{\delta}$.

- (a) (6 points) Compute $f(0, 0)$, $\frac{df(0, 0)}{d(x, y)}$, and $\nabla_{x,y}^2 f(0, 0)$.
- (b) (8 points) Compute all nonzero entries of the third-order derivative tensor $\nabla_{x,y}^3 f(0, 0)$, and use them to derive the third-degree Taylor polynomial $T_3(x, y)$.
- (c) (6 points) Use $T_3(x, y)$ to estimate $f(0.1, 0.2)$, and compare your approximation with the exact value.

Solution.

(a) First,

$$f(0, 0) = 1.$$

The first-order partial derivatives are

$$\frac{\partial f}{\partial x} = e^x + y^2, \quad \frac{\partial f}{\partial y} = 2xy.$$

Hence

$$\frac{df(0, 0)}{d(x, y)} = \begin{bmatrix} 1 & 0 \end{bmatrix}.$$

Next compute the second-order partial derivatives:

$$\frac{\partial^2 f}{\partial x^2} = e^x, \quad \frac{\partial^2 f}{\partial x \partial y} = 2y, \quad \frac{\partial^2 f}{\partial y \partial x} = 2y, \quad \frac{\partial^2 f}{\partial y^2} = 2x.$$

Therefore

$$\nabla_{x,y}^2 f(0, 0) = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}.$$

(b) For the third-order derivatives, we have

$$\frac{\partial^3 f}{\partial x^3} = e^x, \quad \frac{\partial^3 f}{\partial x \partial y^2} = 2, \quad \frac{\partial^3 f}{\partial y \partial x \partial y} = 2, \quad \frac{\partial^3 f}{\partial y^2 \partial x} = 2.$$

Hence the nonzero entries of $\nabla_{x,y}^3 f(0,0)$ are

$$\nabla_{x,y}^3 f(0,0)[1,1,1] = 1, \quad \nabla_{x,y}^3 f(0,0)[1,2,2] = 2, \quad \nabla_{x,y}^3 f(0,0)[2,1,2] = 2, \quad \nabla_{x,y}^3 f(0,0)[2,2,1] = 2.$$

All other entries are zero.

Therefore

$$\nabla_{x,y}^3 f(0,0)\delta^3 = x^3 + 6xy^2.$$

Hence

$$\begin{aligned} T_3(x,y) &= 1 + \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \frac{1}{2} \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \frac{1}{6}(x^3 + 6xy^2) \\ &= 1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + xy^2. \end{aligned}$$

(c) Using the Taylor polynomial,

$$\begin{aligned} T_3(0.1, 0.2) &= 1 + 0.1 + \frac{1}{2}(0.1)^2 + \frac{1}{6}(0.1)^3 + (0.1)(0.2)^2 \\ &= 1.109166\bar{6}. \end{aligned}$$

The exact value is

$$f(0.1, 0.2) = e^{0.1} + 0.1(0.2)^2 = e^{0.1} + 0.004 \approx 1.109170918.$$

Thus the third-degree Taylor approximation is very accurate.

Bonus (5 points)

Let

$$f(\mathbf{x}) = \frac{1}{2}\|\mathbf{A}\mathbf{x} - \mathbf{b}\|_2^2 + \frac{\lambda}{2}\|\mathbf{x}\|_2^2, \quad \mathbf{A} \in \mathbb{R}^{m \times n}, \mathbf{b} \in \mathbb{R}^m, \mathbf{x} \in \mathbb{R}^n,$$

where $\lambda > 0$ is a scalar.

- (a) (1 point) Compute the Hessian matrix of f with respect to $\mathbf{x}$.
- (b) (2 points) Show that this Hessian is positive definite.
- (c) (2 points) Deduce that f has a unique stationary point, and show that it is given by

$$\mathbf{x}^* = (\mathbf{A}^\top \mathbf{A} + \lambda \mathbf{I})^{-1} \mathbf{A}^\top \mathbf{b}.$$

Solution. First expand

$$f(\mathbf{x}) = \frac{1}{2}\mathbf{x}^\top \mathbf{A}^\top \mathbf{A} \mathbf{x} - \mathbf{b}^\top \mathbf{A} \mathbf{x} + \frac{1}{2}\mathbf{b}^\top \mathbf{b} + \frac{\lambda}{2}\mathbf{x}^\top \mathbf{x}.$$

Therefore

$$\frac{df}{d\mathbf{x}} = \mathbf{x}^\top \mathbf{A}^\top \mathbf{A} - \mathbf{b}^\top \mathbf{A} + \lambda \mathbf{x}^\top,$$

and hence

$$\frac{d^2 f}{d\mathbf{x}^2} = \mathbf{A}^\top \mathbf{A} + \lambda \mathbf{I}.$$

To show it is positive definite, let $\mathbf{z} \neq \mathbf{0}$. Then

$$\mathbf{z}^\top \frac{d^2 f}{d\mathbf{x}^2} \mathbf{z} = \mathbf{z}^\top (\mathbf{A}^\top \mathbf{A} + \lambda \mathbf{I}) \mathbf{z} = \|\mathbf{A}\mathbf{z}\|_2^2 + \lambda \|\mathbf{z}\|_2^2 > 0.$$

Therefore the Hessian is positive definite.

A stationary point satisfies

$$\frac{df}{d\mathbf{x}} = \mathbf{0}^\top.$$

Taking transpose gives

$$(\mathbf{A}^\top \mathbf{A} + \lambda \mathbf{I}) \mathbf{x} = \mathbf{A}^\top \mathbf{b}.$$

Hence

$$\mathbf{x}^* = (\mathbf{A}^\top \mathbf{A} + \lambda \mathbf{I})^{-1} \mathbf{A}^\top \mathbf{b}.$$

Since the Hessian is positive definite, this stationary point is unique.